Study 4 Analysis

Table of contents

Setup 1

1. Data integrity checks 2
- 1.1 Do the stored scale scores match the items? . . . 2
- 1.2 Were empathy items reverse-scored? . . . 3
- 1.3 Factor codes, attention checks, scenario comprehension . . . 4
- 1.4 Missing data . . . 4

2. Scale construction and reliability 5

3. Descriptives, randomization, and room to move 7
- 3.1 Cell descriptives . . . 7
- 3.2 Randomization check . . . 7
- 3.3 Room to move . . . 8

4. Manipulation check: post-feedback metaperception (Comment 4) 9
- 4.0 Post-measure diagnostics (v2: run before interpreting 4.1) . . . 9

5. Preregistered confirmatory models 15

6. Robustness 19

7. Pooled stance × feedback models (Comment 3) 21

8. Exploratory: baseline-estimate analyses 24
- 8.1 Linear interaction (as promised in the response letter) . . . 24
- 8.2 Sign-based (kink) test . . . 25

9. Export tables 26

Setup

```
library(readxl)
library(tidyverse)
library(car)
library(emmeans)
library(effectsize)
library(psych)
library(broom)
library(knitr)
```

```r
# Sum-to-zero contrasts so Type III tests and emmeans averages are meaningful
options(contrasts = c("contr.sum", "contr.poly"))

dir_data <- "/Users/yaldasmac/Desktop/Metaperception/study4"
dir_out  <- file.path(dir_data, "reanalysis_output")
dir.create(dir_out, showWarnings = FALSE)

prochoice_raw <- read_excel(file.path(dir_data, "prochoice_intervention.xlsx"))
prolife_raw   <- read_excel(file.path(dir_data, "prolife_intervention.xlsx"))

show_tbl <- function(x, caption = NULL, digits = 3) {
  knitr::kable(x, digits = digits, caption = caption)
}
```

1. Data integrity checks

1.1 Do the stored scale scores match the items?

The pro-choice file contains precomputed scores. Below each one is recomputed from the items and compared with the stored version. **Any score with mismatches must not be used, and any previously reported result built on it must be rechecked.**

```r
chk <- prochoice_raw |>
  mutate(
    morality_recalc = rowMeans(pick(morality_1:morality_6)),
    trust_recalc    = rowMeans(pick(trust_1:trust_6)),
    threat_recalc   = rowMeans(pick(threat_1:threat_6)),
    emp_raw_mean    = rowMeans(pick(emp_1:emp_14))   # all 14 items, NO reverse scoring
  )

compare_scores <- function(stored, recalc, label, recalc_desc) {
  tibble(
    stored_variable = label,
    recomputed_as   = recalc_desc,
    r               = cor(stored, recalc, use = "pairwise.complete.obs"),
    n_mismatch      = sum(abs(stored - recalc) > 0.01, na.rm = TRUE),
    n_total         = sum(!is.na(stored) & !is.na(recalc))
  )
}

bind_rows(
  compare_scores(chk$morality_mean, chk$morality_recalc, "morality_mean", "mean of morality_1-6"),
  compare_scores(chk$trust_score, chk$trust_recalc, "trust_score", "mean of trust_1-6"),
  compare_scores(chk$threat_score, chk$threat_recalc, "threat_score", "mean of threat_1-6"),
  compare_scores(chk$perspective_taking, chk$emp_raw_mean, "perspective_taking",
                 "mean of emp_1-14, no reverse scoring")
) |> show_tbl("Stored vs. recomputed scores (pro-choice file)")
```

Table 1: Stored vs. recomputed scores (pro-choice file)

stored_variable	recomputed_as	r	n_mismatch	n_total
morality_mean	mean of morality_1–6	1.000	0	604
trust_score	mean of trust_1–6	0.916	103	604
threat_score	mean of threat_1–6	0.906	129	604
perspective_taking	mean of emp_1–14, no reverse scoring	1.000	1	604

If `trust_score` or `threat_score` mismatch, check whether they were swapped or shifted between rows:

```
chk |>
  select(trust_score, threat_score, trust_recalc, threat_recalc) |>
  cor(use = "pairwise.complete.obs") |>
  round(3) |>
  show_tbl("Correlations among stored and recomputed trust/threat")
```

Table 2: Correlations among stored and recomputed trust/threat

	trust_score	threat_score	trust_recalc	threat_recalc
trust_score	1.000	-0.713	0.916	-0.654
threat_score	-0.713	1.000	-0.668	0.906
trust_recalc	0.916	-0.668	1.000	-0.728
threat_recalc	-0.654	0.906	-0.728	1.000

1.2 Were empathy items reverse-scored?

The IRI items keyed negatively are emp_2, emp_4, and emp_5 (Empathic Concern) and emp_8 and emp_11 (Perspective Taking). This assumes the Qualtrics item order matches the SI.

If the data are stored raw, those items should correlate **negatively** with the positively keyed items of their subscale.

```
ec_pos <- c("emp_1", "emp_3", "emp_6", "emp_7")
pt_pos <- c("emp_9", "emp_10", "emp_12", "emp_13", "emp_14")

reverse_diag <- function(d, stance_label) {
  ec_key <- rowMeans(d[ec_pos], na.rm = TRUE)
  pt_key <- rowMeans(d[pt_pos], na.rm = TRUE)
  tibble(
    stance   = stance_label,
    item     = paste0("emp_", 1:14),
    subscale = rep(c("EC", "PT"), each = 7),
    r_with_positive_items = map2_dbl(item, subscale, \(it, s)
      cor(d[[it]], if (s == "EC") ec_key else pt_key, use = "pairwise.complete.obs"))
  )
}

bind_rows(reverse_diag(prochoice_raw, "Pro-choice"),
          reverse_diag(prolife_raw, "Pro-life")) |>
  pivot_wider(names_from = stance, values_from = r_with_positive_items) |>
  show_tbl("Item correlations with positively keyed items (negative = item is stored raw and must be
   ↪  reversed)")
```

Table 3: Item correlations with positively keyed items (negative = item is stored raw and must be reversed)

item	subscale	Pro-choice	Pro-life
emp_1	EC	0.886	0.856
emp_2	EC	0.461	0.359
emp_3	EC	0.849	0.844
emp_4	EC	0.509	0.405
emp_5	EC	0.416	0.355
emp_6	EC	0.865	0.832
emp_7	EC	0.868	0.841
emp_8	PT	0.388	0.303
emp_9	PT	0.791	0.828
emp_10	PT	0.771	0.843
emp_11	PT	0.340	0.222
emp_12	PT	0.769	0.800
emp_13	PT	0.850	0.844
emp_14	PT	0.870	0.874

1.3 Factor codes, attention checks, scenario comprehension

```
prochoice_raw |> count(intervention, intervention_factor) |>
  show_tbl("Pro-choice: numeric vs. stored feedback labels (1 = Better, 2 = Worse, 3 = Control)")
```

Table 4: Pro-choice: numeric vs. stored feedback labels (1 = Better, 2 = Worse, 3 = Control)

intervention	intervention_factor	n
1	Better	200
2	Worse	214
3	Control	190

```
prochoice_raw |> count(group, group_factor) |>
  show_tbl("Pro-choice: numeric vs. stored moral labels (1 = Individualizing, 2 = Binding)")
```

Table 5: Pro-choice: numeric vs. stored moral labels (1 = Individualizing, 2 = Binding)

group	group_factor	n
1	Individualizing	309
2	Binding	295

```
att_tab <- function(d, stance_label) {
  d |>
    mutate(stance = stance_label,
           feedback_code = intervention,
           attention_response = coalesce(high_attention, low_attention, control_attention)) |>
    count(stance, feedback_code, attention_response)
}
bind_rows(att_tab(prochoice_raw, "Pro-choice"), att_tab(prolife_raw, "Pro-life")) |>
  show_tbl("Feedback-identification responses by condition (1 = Better, 2 = Worse, 3 = Control)")
```

Table 6: Feedback-identification responses by condition (1 = Better, 2 = Worse, 3 = Control)

stance	feedback_code	attention_response	n
Pro-choice	1	Prolife Americans rated Sara as more moral than prochoice participants expected.	200
Pro-choice	2	Prolife Americans rated Sara as less moral than prochoice participants expected.	214
Pro-choice	3	Prolife Americans rated Sara almost the same as prochoice participants expected.	190
Pro-life	1	Prochoice Americans rated Emily as more moral than prolife participants expected.	204
Pro-life	2	Prochoice Americans rated Emily as less moral than prolife participants expected.	205
Pro-life	3	Prochoice Americans rated Emily almost the same as prolife participants expected.	191

```
bind_rows(
  prochoice_raw |> count(scenario_attention_check) |> mutate(stance = "Pro-choice"),
  prolife_raw   |> count(scenario_attention_check) |> mutate(stance = "Pro-life")
) |> show_tbl("Scenario comprehension responses")
```

Table 7: Scenario comprehension responses

scenario_attention_check	n	stance
She shares her story at rallies to emphasize the importance of women,Äôs autonomy.	604	Pro-choice
Emily supports access to abortion and advocates for prochoice policies.	600	Pro-life

1.4 Missing data

```
key_items <- c(paste0("morality_", 1:6), paste0("post_morality_", 1:6),
               paste0("trust_", 1:6), paste0("threat_", 1:6),
               paste0("emp_", 1:14), "ideology")
bind_rows(
  prochoice_raw |> summarise(across(all_of(key_items), ~ sum(is.na(.x)))) |> mutate(stance = "Pro-choice"),
  prolife_raw   |> summarise(across(all_of(key_items), ~ sum(is.na(.x)))) |> mutate(stance = "Pro-life")
) |>
  pivot_longer(-stance, names_to = "item", values_to = "n_missing") |>
  filter(n_missing > 0) |>
  show_tbl("Items with missing values (empty table = none)")
```

Table 8: Items with missing values (empty table = none)

stance	item	n_missing

2. Scale construction and reliability

```
REVERSE_EMPATHY <- FALSE   # items already reverse-coded at export (see 1.2)
emp_rev_items <- if (REVERSE_EMPATHY) c("emp_2", "emp_4", "emp_5", "emp_8", "emp_11") else character(0)

build_scales <- function(d, stance_label) {
  d |>
    rename(any_of(c(income = "Q25"))) |>
    select(-any_of(c("morality_mean", "threat_score", "trust_score",
                     "perspective_taking", "intervention_factor", "group_factor"))) |>
    mutate(
      stance   = stance_label,
      moral    = factor(group, levels = 1:2, labels = c("Individualizing", "Binding")),
      feedback = factor(intervention, levels = 1:3, labels = c("Better", "Worse", "Control")),
      # Creates emp_k_r for every item; reversed only if REVERSE_EMPATHY is TRUE
      across(num_range("emp_", 1:14), ~ if (cur_column() %in% emp_rev_items) 6 - .x else .x,
             .names = "{.col}_r"),
      meta_pre    = rowMeans(pick(morality_1:morality_6)),
      meta_post   = rowMeans(pick(post_morality_1:post_morality_6)),
      meta_change = meta_post - meta_pre,
      trust       = rowMeans(pick(trust_1:trust_6)),
      threat      = rowMeans(pick(threat_1:threat_6)),
      threat_realistic = rowMeans(pick(threat_1:threat_3)),
      threat_symbolic  = rowMeans(pick(threat_4:threat_6)),
      EC = rowMeans(pick(emp_1, emp_2_r, emp_3, emp_4_r, emp_5_r, emp_6, emp_7)),
      PT = rowMeans(pick(emp_8_r, emp_9, emp_10, emp_11_r, emp_12, emp_13, emp_14)),
      empathy = (EC + PT) / 2
    )
}

s4 <- bind_rows(
  build_scales(prochoice_raw, "Pro-choice"),
  build_scales(prolife_raw, "Pro-life")
) |>
  mutate(stance = factor(stance, levels = c("Pro-choice", "Pro-life"))) |>
  # Centering within stance (moderators, covariates)
  group_by(stance) |>
  mutate(PT_c      = PT - mean(PT, na.rm = TRUE),
         empathy_c = empathy - mean(empathy, na.rm = TRUE),
         ideo_c    = ideology - mean(ideology, na.rm = TRUE)) |>
  # v2 fix: pre-feedback metaperception centered within STANCE only (not stance x moral).
  # Centering within cells left the moral/stance:moral terms unadjusted for baseline,
  # which produced the inflated F values (e.g., 2210, 3280) in v1. Feedback contrasts are unaffected.
  mutate(meta_pre_c = meta_pre - mean(meta_pre, na.rm = TRUE)) |>
  ungroup()
```

```
s4 |> count(stance, moral, feedback) |>
  pivot_wider(names_from = feedback, values_from = n) |>
  show_tbl("Cell sizes (should match SI Tables S-cells)")
```

Table 9: Cell sizes (should match SI Tables S-cells)

stance	moral	Better	Worse	Control
Pro-choice	Individualizing	104	110	95
Pro-choice	Binding	96	104	95
Pro-life	Individualizing	101	102	94
Pro-life	Binding	103	103	97

```
alpha_of <- function(d, items) {
  psych::alpha(d[items], warnings = FALSE)$total$raw_alpha
}

rel <- s4 |>
  group_by(stance, moral) |>
  group_modify(\(d, k) tibble(
    meta_pre  = alpha_of(d, paste0("morality_", 1:6)),
    meta_post = alpha_of(d, paste0("post_morality_", 1:6))
  )) |> ungroup()

rel_dv <- s4 |>
  group_by(stance) |>
  group_modify(\(d, k) tibble(
    trust  = alpha_of(d, paste0("trust_", 1:6)),
    threat = alpha_of(d, paste0("threat_", 1:6)),
    EC     = alpha_of(d, c("emp_1", "emp_2_r", "emp_3", "emp_4_r", "emp_5_r", "emp_6", "emp_7")),
    PT     = alpha_of(d, c("emp_8_r", "emp_9", "emp_10", "emp_11_r", "emp_12", "emp_13", "emp_14"))
  )) |> ungroup()

show_tbl(rel, "Cronbach's alpha: metaperception (by stance x moral condition)")
```

Table 10: Cronbach's alpha: metaperception (by stance x moral condition)

stance	moral	meta_pre	meta_post
Pro-choice	Individualizing	0.785	0.825
Pro-choice	Binding	0.915	0.919
Pro-life	Individualizing	0.691	0.722
Pro-life	Binding	0.873	0.818

```
show_tbl(rel_dv, "Cronbach's alpha: outcomes and moderators (by stance)")
```

Table 11: Cronbach's alpha: outcomes and moderators (by stance)

stance	trust	threat	EC	PT
Pro-choice	0.940	0.924	0.878	0.833
Pro-life	0.936	0.899	0.842	0.826

```
s4 |> group_by(stance) |>
  summarise(r_realistic_symbolic = cor(threat_realistic, threat_symbolic),
            r_EC_PT = cor(EC, PT)) |>
  show_tbl("Subscale intercorrelations")
```

Table 12: Subscale intercorrelations

stance	r_realistic_symbolic	r_EC_PT
Pro-choice	0.793	0.543
Pro-life	0.588	0.523

3. Descriptives, randomization, and room to move

3.1 Cell descriptives

```
cell_desc <- s4 |>
  group_by(stance, moral, feedback) |>
  summarise(
    n = n(),
    pre_M = mean(meta_pre), pre_SD = sd(meta_pre),
    post_M = mean(meta_post), post_SD = sd(meta_post),
    change_M = mean(meta_change),
    trust_M = mean(trust), trust_SD = sd(trust),
    threat_M = mean(threat), threat_SD = sd(threat),
    .groups = "drop"
  )
show_tbl(cell_desc, "Descriptives by stance x moral x feedback", digits = 2)
```

Table 13: Descriptives by stance x moral x feedback

stance	moral	feedback	n	pre_M	pre_SD	post_M	post_SD	change_M	trust_M	trust_SD	threat_M	threat_SD
Pro-choice	Individualizing	Better	104	2.77	1.02	3.49	0.84	0.72	5.22	0.62	2.65	0.87
Pro-choice	Individualizing	Worse	110	2.83	0.90	2.52	0.73	-0.31	3.97	1.56	4.39	1.65
Pro-choice	Individualizing	Control	95	2.77	0.95	2.82	1.01	0.05	3.99	1.54	4.13	1.70
Pro-choice	Binding	Better	96	1.91	1.00	2.60	0.93	0.68	5.04	0.77	2.93	0.99
Pro-choice	Binding	Worse	104	1.96	1.00	1.65	0.87	-0.31	3.97	1.54	4.26	1.56
Pro-choice	Binding	Control	95	1.75	0.80	1.79	0.79	0.04	4.35	1.45	3.89	1.61
Pro-life	Individualizing	Better	101	2.86	0.79	3.20	0.87	0.34	4.77	1.29	3.60	1.45
Pro-life	Individualizing	Worse	102	2.78	0.89	2.55	0.81	-0.23	4.05	1.51	4.37	1.53
Pro-life	Individualizing	Control	94	2.76	0.84	2.76	0.78	0.01	4.31	1.32	4.13	1.43
Pro-life	Binding	Better	103	3.50	0.81	3.68	0.77	0.18	4.45	1.53	4.03	1.48
Pro-life	Binding	Worse	103	3.54	0.66	3.37	0.64	-0.17	3.93	1.36	4.48	1.33
Pro-life	Binding	Control	97	3.58	0.85	3.57	0.85	-0.01	4.42	1.31	4.05	1.40

3.2 Randomization check

These variables were measured before feedback, or are traits, so they should **not** differ by feedback condition.

```
balance_vars <- c("meta_pre", "ideology", "PT", "EC", "age")

balance <- s4 |>
  group_by(stance) |>
  group_modify(\(d, k) map_dfr(balance_vars, \(v) {
    a <- car::Anova(lm(reformulate("feedback * moral", v), data = d), type = 3)
    tibble(variable = v,
           F_feedback = a["feedback", "F value"], p_feedback = a["feedback", "Pr(>F)"],
           F_fb_x_moral = a["feedback:moral", "F value"], p_fb_x_moral = a["feedback:moral", "Pr(>F)"])
  })) |> ungroup()
show_tbl(balance, "Randomization check (Type III F tests)")
```

Table 14: Randomization check (Type III F tests)

stance	variable	F_feedback	p_feedback	F_fb_x_moral	p_fb_x_moral
Pro-choice	meta_pre	0.966	0.381	0.451	0.638
Pro-choice	ideology	1.940	0.145	0.951	0.387
Pro-choice	PT	0.397	0.673	1.001	0.368
Pro-choice	EC	0.248	0.780	0.380	0.684
Pro-choice	age	0.787	0.456	0.872	0.419
Pro-life	meta_pre	0.034	0.967	0.672	0.511
Pro-life	ideology	1.043	0.353	0.749	0.473
Pro-life	PT	0.041	0.960	0.967	0.381
Pro-life	EC	0.253	0.777	0.149	0.861
Pro-life	age	1.142	0.320	0.838	0.433

3.3 Room to move

The feedback showed fixed values: Better = 3.8, Control = 2.8, Worse = 1.2. Pre-feedback estimates show how much room each group had to move. **Pro-life pre-feedback means above 2.8 imply that "control" was effectively a downward message for them.**

```
s4 |>
  group_by(stance, moral) |>
  summarise(
    pre_M = mean(meta_pre), pre_SD = sd(meta_pre),
    pct_above_3.8 = mean(meta_pre > 3.8) * 100,
    pct_above_2.8 = mean(meta_pre > 2.8) * 100,
    pct_below_1.2 = mean(meta_pre < 1.2) * 100,
    .groups = "drop"
  ) |>
  show_tbl("Pre-feedback metaperception relative to displayed values", digits = 2)
```

Table 15: Pre-feedback metaperception relative to displayed values

stance	moral	pre_M	pre_SD	pct_above_3.8	pct_above_2.8	pct_below_1.2
Pro-choice	Individualizing	2.79	0.95	15.53	51.46	6.47
Pro-choice	Binding	1.88	0.94	5.76	18.64	31.53
Pro-life	Individualizing	2.80	0.84	13.47	49.83	3.37
Pro-life	Binding	3.54	0.77	47.52	85.81	0.99

```
s4 |>
  group_by(stance, feedback) |>
  summarise(trust_pct_at_7 = mean(trust == 7) * 100,
            trust_pct_at_1 = mean(trust == 1) * 100,
            threat_pct_at_7 = mean(threat == 7) * 100,
            threat_pct_at_1 = mean(threat == 1) * 100,
            .groups = "drop") |>
  show_tbl("Ceiling/floor on outcomes (% at scale endpoints)", digits = 1)
```

Table 16: Ceiling/floor on outcomes (% at scale endpoints)

stance	feedback	trust_pct_at_7	trust_pct_at_1	threat_pct_at_7	threat_pct_at_1
Pro-choice	Better	0.5	0.0	0.0	4.0
Pro-choice	Worse	1.4	5.1	6.1	6.1
Pro-choice	Control	1.1	4.7	3.7	8.9
Pro-life	Better	5.4	2.0	2.0	9.3
Pro-life	Worse	1.5	3.9	4.4	3.9
Pro-life	Control	0.5	1.0	1.6	3.1

```r
ggplot(s4, aes(meta_pre, fill = stance)) +
  geom_histogram(binwidth = 1/6, alpha = .6, position = "identity") +
  geom_vline(xintercept = c(1.2, 2.8, 3.8), linetype = "dashed") +
  facet_wrap(~ moral) +
  labs(x = "Pre-feedback metaperception (1-5)", y = "Count",
       title = "Pre-feedback estimates; dashed lines = values shown (Worse, Control, Better)") +
  theme_minimal()
```

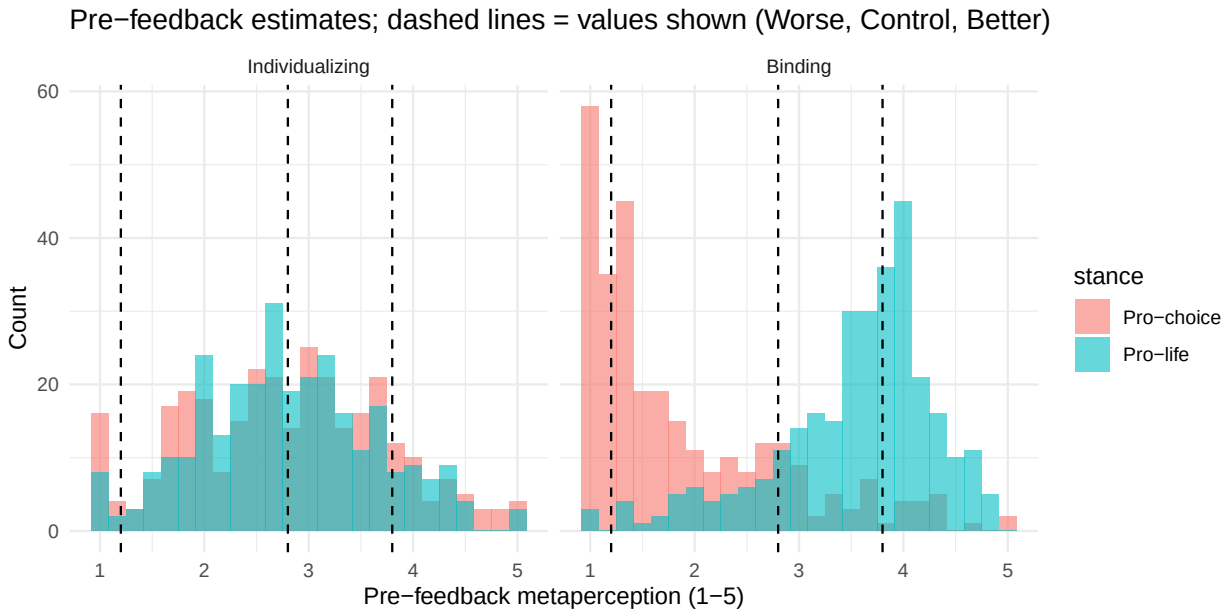


Figure 1

4. Manipulation check: post-feedback metaperception (Comment 4)

Pre-specified success criteria. Each must hold within each stance group:

1. A significant omnibus feedback effect on post-feedback metaperception, adjusting for pre-feedback metaperception.
2. **Better > Worse**, significant within each moral condition. Better > Control and Control > Worse are welcome but not required.
3. Movement toward the displayed value: change from pre-feedback is positive in Better, negative in Worse, and near zero in Control. For pro-life participants, a small decrease in Control is expected if their pre-feedback means exceed 2.8.

Caveat for the write-up: the post-measure was administered *after* trust and threat. It is a manipulation check, not a mediator.

4.0 Post-measure diagnostics (v2: run before interpreting 4.1)

v1 raised two concerns:

- **Anomalous cell.** In the pro-choice × individualizing × Worse cell, metaperceptions *rose* by 0.47 after worse-than-expected feedback, even though trust fell in that cell.

- **Unusually regular change.** The SD of change is very small in several cells (for example, pro-life × binding × Better has mean +0.23 and SD ≈ 0.14, so almost everyone moved up by one or two item-points and almost no one moved down).

The code below describes the change distributions item by item. **Compare these columns against a fresh raw Qualtrics export before using them.**

```
change_dist <- s4 |>
  mutate(item_ups   = rowSums(pick(post_morality_1:post_morality_6) > pick(morality_1:morality_6)),
         item_downs = rowSums(pick(post_morality_1:post_morality_6) < pick(morality_1:morality_6))) |>
  group_by(stance, moral, feedback) |>
  summarise(n = n(),
            change_SD        = sd(meta_change),
            pct_identical    = mean(meta_change == 0) * 100,
            pct_any_up       = mean(item_ups > 0) * 100,
            pct_any_down     = mean(item_downs > 0) * 100,
            pct_up_and_down  = mean(item_ups > 0 & item_downs > 0) * 100,
            mean_items_up    = mean(item_ups),
            mean_items_down  = mean(item_downs),
            r_pre_post       = cor(meta_pre, meta_post),
            .groups = "drop")
show_tbl(change_dist, "Distribution of pre-to-post change by cell", digits = 2)
```

Table 17: Distribution of pre-to-post change by cell

stance	moral	feedback	n	change_SD	pct_identical	pct_any_up	pct_any_down	pct_up_and_down	mean_items_up	mean_items_down	r_pre_post
Pro-choice	Individualizing	Better	104	0.32	3.85	96.15	0.96	0.96	3.81	0.01	0.96
Pro-choice	Individualizing	Worse	110	0.31	14.55	42.73	86.36	38.18	0.52	2.11	0.95
Pro-choice	Individualizing	Control	95	0.32	24.21	66.32	57.89	32.63	1.19	0.93	0.95
Pro-choice	Binding	Better	96	0.34	3.12	97.92	9.38	9.38	3.62	0.09	0.94
Pro-choice	Binding	Worse	104	0.29	26.92	1.92	73.08	1.92	0.03	1.79	0.96
Pro-choice	Binding	Control	95	0.29	33.68	50.53	53.68	26.32	1.01	0.78	0.93
Pro-life	Individualizing	Better	101	0.51	12.87	76.24	35.64	16.83	2.41	0.64	0.81
Pro-life	Individualizing	Worse	102	0.22	22.55	18.63	76.47	13.73	0.19	1.54	0.97
Pro-life	Individualizing	Control	94	0.18	12.77	52.13	47.87	6.38	0.57	0.53	0.98
Pro-life	Binding	Better	103	0.24	20.39	86.41	46.60	38.83	1.71	0.64	0.95
Pro-life	Binding	Worse	103	0.20	31.07	8.74	67.96	3.88	0.09	1.07	0.96
Pro-life	Binding	Control	97	0.18	14.43	46.39	49.48	5.15	0.51	0.55	0.98

```
# Item-level change: does any single item drive a cell's shift?
item_change <- map_dfr(1:6, \(i) s4 |>
  mutate(d = .data[[paste0("post_morality_", i)]] - .data[[paste0("morality_", i)]]) |>
  group_by(stance, moral, feedback) |>
  summarise(item = i, mean_change = mean(d), pct_up = mean(d > 0) * 100,
            pct_down = mean(d < 0) * 100, .groups = "drop"))
item_change |>
  select(stance, moral, feedback, item, mean_change) |>
  pivot_wider(names_from = item, values_from = mean_change, names_prefix = "item_") |>
  show_tbl("Mean item-level change (post - pre) by cell", digits = 2)
```

Table 18: Mean item-level change (post - pre) by cell

stance	moral	feedback	item_1	item_2	item_3	item_4	item_5	item_6
Pro-choice	Individualizing	Better	0.61	0.74	0.62	0.75	0.87	0.75
Pro-choice	Individualizing	Worse	-0.32	-0.30	-0.43	-0.37	-0.11	-0.33
Pro-choice	Individualizing	Control	-0.04	0.07	-0.08	0.02	0.16	0.16
Pro-choice	Binding	Better	0.74	0.69	0.67	0.66	0.73	0.61
Pro-choice	Binding	Worse	-0.25	-0.50	-0.25	-0.35	-0.14	-0.37
Pro-choice	Binding	Control	-0.04	-0.12	0.07	0.19	0.15	-0.03
Pro-life	Individualizing	Better	0.33	0.29	0.33	0.39	0.41	0.30
Pro-life	Individualizing	Worse	-0.26	-0.14	-0.20	-0.21	-0.25	-0.31
Pro-life	Individualizing	Control	0.03	-0.04	0.04	0.02	-0.01	0.00
Pro-life	Binding	Better	0.17	0.21	0.20	0.09	0.19	0.20
Pro-life	Binding	Worse	-0.19	-0.17	-0.20	-0.12	-0.17	-0.20
Pro-life	Binding	Control	-0.01	0.04	0.00	-0.02	-0.02	-0.03

```r
# Size of the change (in item-points summed over the 6 items) for every participant
s4 |>
  mutate(points = round(meta_change * 6)) |>
  count(stance, moral, feedback, points) |>
  pivot_wider(names_from = points, values_from = n, values_fill = 0, names_sort = TRUE) |>
  show_tbl("Distribution of summed item-point change per participant")
```

Table 19: Distribution of summed item-point change per participant

stance	moral	feedback	-8	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7	8	9	10
Pro-choice	Individualizing	Better	0	0	0	0	0	0	0	4	5	7	16	21	24	16	7	3	0	1
Pro-choice	Individualizing	Worse	1	1	9	11	13	24	25	16	9	1	0	0	0	0	0	0	0	0
Pro-choice	Individualizing	Control	0	0	0	1	5	12	14	23	16	8	13	2	1	0	0	0	0	0
Pro-choice	Binding	Better	0	0	0	0	0	0	0	3	6	13	17	19	14	10	10	2	2	0
Pro-choice	Binding	Worse	0	3	8	11	11	17	25	28	1	0	0	0	0	0	0	0	0	0
Pro-choice	Binding	Control	0	0	0	0	5	8	19	32	7	11	11	1	1	0	0	0	0	0
Pro-life	Individualizing	Better	0	0	2	2	4	4	9	13	7	11	17	13	8	5	2	2	0	2
Pro-life	Individualizing	Worse	0	0	1	7	10	26	30	23	5	0	0	0	0	0	0	0	0	0
Pro-life	Individualizing	Control	0	0	0	0	0	5	34	12	38	5	0	0	0	0	0	0	0	0
Pro-life	Binding	Better	0	0	0	0	1	2	12	21	26	24	13	4	0	0	0	0	0	0
Pro-life	Binding	Worse	0	0	0	4	9	17	36	32	5	0	0	0	0	0	0	0	0	0
Pro-life	Binding	Control	0	0	0	0	0	5	38	14	36	4	0	0	0	0	0	0	0	0

```r
ggplot(s4, aes(meta_change)) +
  geom_histogram(binwidth = 1/6, boundary = -1/12) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  facet_grid(stance + moral ~ feedback) +
  labs(x = "Post - pre metaperception", y = "Count",
       title = "Pre-to-post change by cell") +
  theme_minimal()
```

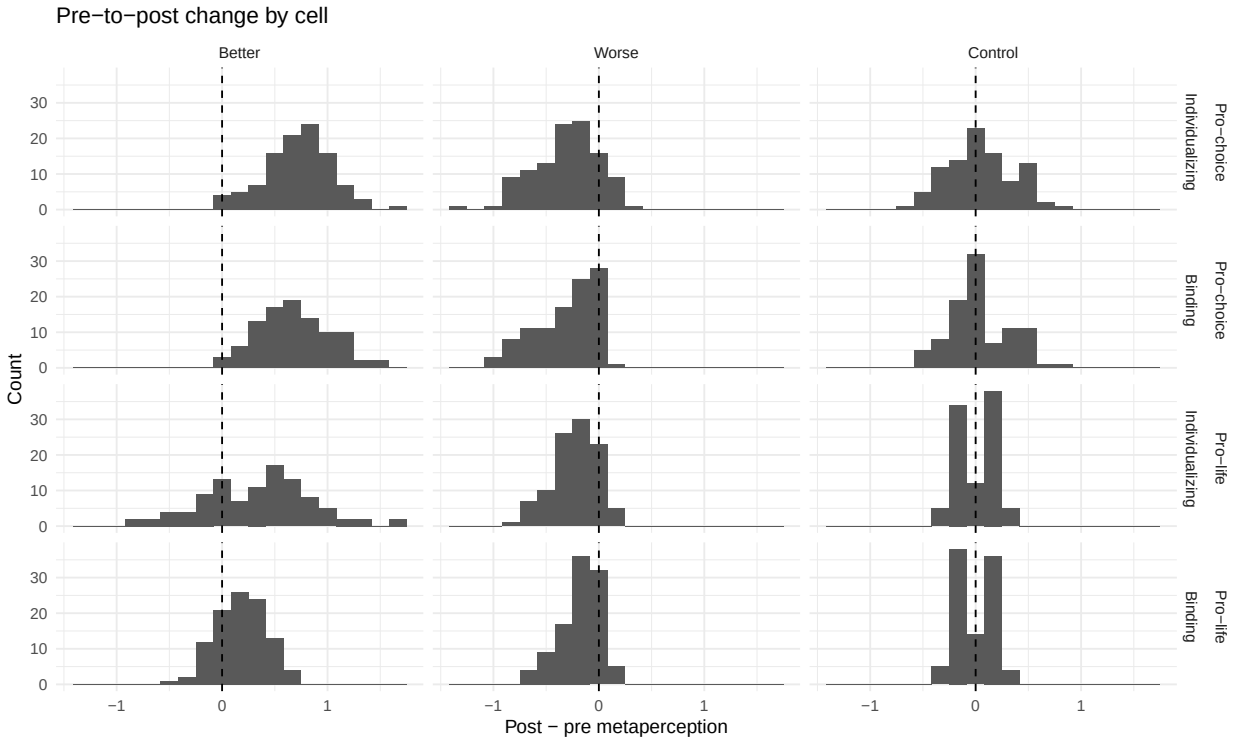


Figure 2

```
anova_table <- function(fit) {
  a  <- car::Anova(fit, type = 3)
  es <- effectsize::eta_squared(a, partial = TRUE, ci = .95, verbose = FALSE) |>
    as_tibble() |>
    select(term = Parameter, eta2_p = Eta2_partial, eta2_CI_low = CI_low, eta2_CI_high = CI_high)
  a |>
    as.data.frame() |>
    rownames_to_column("term") |>
    filter(!term %in% c("(Intercept)", "Residuals")) |>
    transmute(term, df1 = Df, df2 = fit$df.residual, F = `F value`, p = `Pr(>F)`) |>
    left_join(es, by = "term")
}

pairs_table <- function(fit, spec, adjust = "tukey") {
  summary(pairs(emmeans(fit, spec), adjust = adjust), infer = TRUE) |> as_tibble()
}

emm_table <- function(fit, spec) {
  summary(emmeans(fit, spec)) |> as_tibble()
}

by_stance <- split(s4, s4$stance)
```

```
manip_fits <- map(by_stance, \(d) lm(meta_post ~ feedback * moral + meta_pre_c, data = d))

manip_anova <- imap_dfr(manip_fits, \(f, s) anova_table(f) |> mutate(stance = s, .before = 1))
show_tbl(manip_anova, "Criterion 1: ANCOVA on post-feedback metaperception (Type III)")
```

Table 20: Criterion 1: ANCOVA on post-feedback metaperception (Type III)

stance	term	df1	df2	F	p	eta2_p	eta2_CI_low	eta2_CI_high
Pro-choice	feedback	2	597	672.841	0.000	0.693	0.663	1
Pro-choice	moral	1	597	31.725	0.000	0.050	0.026	1
Pro-choice	meta_pre_c	1	597	4998.812	0.000	0.893	0.882	1
Pro-choice	feedback:moral	2	597	0.266	0.767	0.001	0.000	1
Pro-life	feedback	2	593	146.823	0.000	0.331	0.282	1
Pro-life	moral	1	593	0.943	0.332	0.002	0.000	1
Pro-life	meta_pre_c	1	593	4360.232	0.000	0.880	0.868	1
Pro-life	feedback:moral	2	593	9.190	0.000	0.030	0.010	1

```
manip_emm <- imap_dfr(manip_fits, \(f, s) emm_table(f, ~ feedback | moral) |> mutate(stance = s, .before = 1))
show_tbl(manip_emm, "Adjusted post-feedback metaperception (at mean pre-feedback score)")
```

Table 21: Adjusted post-feedback metaperception (at mean pre-feedback score)

stance	feedback	moral	emmean	SE	df	lower.CL	upper.CL
Pro-choice	Better	Individualizing	3.126	0.028	597	3.071	3.181
Pro-choice	Worse	Individualizing	2.103	0.028	597	2.049	2.157
Pro-choice	Control	Individualizing	2.452	0.029	597	2.394	2.510
Pro-choice	Better	Binding	2.968	0.029	597	2.910	3.025
Pro-choice	Worse	Binding	1.982	0.028	597	1.927	2.038
Pro-choice	Control	Binding	2.300	0.030	597	2.241	2.358
Pro-life	Better	Individualizing	3.485	0.028	593	3.431	3.539
Pro-life	Worse	Individualizing	2.911	0.028	593	2.857	2.965
Pro-life	Control	Individualizing	3.145	0.029	593	3.088	3.201
Pro-life	Better	Binding	3.380	0.027	593	3.326	3.433
Pro-life	Worse	Binding	3.030	0.027	593	2.977	3.084
Pro-life	Control	Binding	3.202	0.028	593	3.146	3.257

```
manip_pairs <- imap_dfr(manip_fits, \(f, s) pairs_table(f, ~ feedback | moral) |> mutate(stance = s, .before =
↪  1))
show_tbl(manip_pairs, "Criterion 2: pairwise contrasts within moral condition (Tukey)")
```

Table 22: Criterion 2: pairwise contrasts within moral condition (Tukey)

stance	contrast	moral	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
Pro-choice	Better - Worse	Individualizing	1.023	0.039	597	0.932	1.113	26.507	0
Pro-choice	Better - Control	Individualizing	0.674	0.040	597	0.580	0.768	16.832	0
Pro-choice	Worse - Control	Individualizing	-0.349	0.040	597	-0.442	-0.256	-8.831	0
Pro-choice	Better - Worse	Binding	0.985	0.040	597	0.892	1.079	24.677	0
Pro-choice	Better - Control	Binding	0.668	0.041	597	0.572	0.764	16.346	0
Pro-choice	Worse - Control	Binding	-0.317	0.040	597	-0.412	-0.223	-7.909	0
Pro-life	Better - Worse	Individualizing	0.574	0.038	593	0.484	0.664	14.973	0
Pro-life	Better - Control	Individualizing	0.340	0.039	593	0.248	0.432	8.688	0
Pro-life	Worse - Control	Individualizing	-0.234	0.039	593	-0.326	-0.142	-5.992	0
Pro-life	Better - Worse	Binding	0.349	0.038	593	0.260	0.439	9.179	0
Pro-life	Better - Control	Binding	0.178	0.039	593	0.087	0.269	4.603	0
Pro-life	Worse - Control	Binding	-0.171	0.039	593	-0.262	-0.081	-4.435	0

```
manip_pairs_collapsed <- imap_dfr(manip_fits, \(f, s) pairs_table(f, ~ feedback) |> mutate(stance = s, .before
↪  = 1))
show_tbl(manip_pairs_collapsed, "Pairwise contrasts averaged over moral condition (Tukey)")
```

Table 23: Pairwise contrasts averaged over moral condition (Tukey)

stance	contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
Pro-choice	Better - Worse	1.004	0.028	597	0.939	1.069	36.161	0
Pro-choice	Better - Control	0.671	0.029	597	0.604	0.738	23.455	0
Pro-choice	Worse - Control	-0.333	0.028	597	-0.399	-0.267	-11.826	0
Pro-life	Better - Worse	0.462	0.027	593	0.398	0.525	17.097	0
Pro-life	Better - Control	0.259	0.027	593	0.194	0.324	9.422	0
Pro-life	Worse - Control	-0.203	0.027	593	-0.267	-0.138	-7.379	0

```
manip_change <- s4 |>
  group_by(stance, moral, feedback) |>
  summarise(
    n = n(),
    pre_M = mean(meta_pre), post_M = mean(meta_post),
    change_M = mean(meta_change),
    change_SE = sd(meta_change) / sqrt(n),
    t = change_M / change_SE,
    p = 2 * pt(-abs(t), df = n - 1),
    .groups = "drop"
  )
show_tbl(manip_change, "Criterion 3: within-cell change from pre to post (one-sample t vs. 0)")
```

Table 24: Criterion 3: within-cell change from pre to post (one-sample t vs. 0)

stance	moral	feedback	n	pre_M	post_M	change_M	change_SE	t	p
Pro-choice	Individualizing	Better	104	2.772	3.494	0.721	0.031	23.068	0.000
Pro-choice	Individualizing	Worse	110	2.826	2.517	-0.309	0.029	-10.505	0.000
Pro-choice	Individualizing	Control	95	2.772	2.819	0.047	0.032	1.462	0.147
Pro-choice	Binding	Better	96	1.915	2.597	0.682	0.035	19.642	0.000
Pro-choice	Binding	Worse	104	1.960	1.651	-0.309	0.029	-10.757	0.000
Pro-choice	Binding	Control	95	1.753	1.789	0.037	0.030	1.243	0.217
Pro-life	Individualizing	Better	101	2.861	3.200	0.338	0.051	6.636	0.000
Pro-life	Individualizing	Worse	102	2.779	2.551	-0.229	0.022	-10.566	0.000
Pro-life	Individualizing	Control	94	2.757	2.764	0.007	0.019	0.376	0.708
Pro-life	Binding	Better	103	3.500	3.678	0.178	0.024	7.474	0.000
Pro-life	Binding	Worse	103	3.540	3.366	-0.175	0.019	-9.059	0.000
Pro-life	Binding	Control	97	3.581	3.574	-0.007	0.018	-0.380	0.705

```r
# Does the manipulation move beliefs equally in both stance groups?
manip_pooled <- lm(meta_post ~ feedback * stance * moral + meta_pre_c, data = s4)
show_tbl(anova_table(manip_pooled), "Pooled ANCOVA on post-feedback metaperception")
```

Table 25: Pooled ANCOVA on post-feedback metaperception

term	df1	df2	F	p	eta2_p	eta2_CI_low	eta2_CI_high
feedback	2	1191	727.236	0.000	0.550	0.522	1
stance	1	1191	1914.954	0.000	0.617	0.592	1
moral	1	1191	5.708	0.017	0.005	0.000	1
meta_pre_c	1	1191	9313.988	0.000	0.887	0.879	1
feedback:stance	2	1191	105.077	0.000	0.150	0.120	1
feedback:moral	2	1191	6.056	0.002	0.010	0.002	1
stance:moral	1	1191	22.932	0.000	0.019	0.008	1
feedback:stance:moral	2	1191	3.412	0.033	0.006	0.000	1

```r
manip_int <- summary(
  contrast(emmeans(manip_pooled, ~ feedback * stance),
           interaction = c("pairwise", "pairwise"), adjust = "holm"),
  infer = TRUE) |> as_tibble()
show_tbl(manip_int,
         "Stance x feedback interaction contrasts on post-metaperception (positive = larger contrast among
         ↪  pro-choice)")
```

Table 26: Stance x feedback interaction contrasts on post-metaperception (positive = larger contrast among pro-choice)

feedback_pairwise	stance_pairwise	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
Better - Worse	(Pro-choice) - (Pro-life)	0.543	0.039	1191	0.450	0.636	13.971	0.000
Better - Control	(Pro-choice) - (Pro-life)	0.410	0.040	1191	0.314	0.505	10.298	0.000
Worse - Control	(Pro-choice) - (Pro-life)	-0.133	0.039	1191	-0.228	-0.039	-3.374	0.001

```r
# Robustness: allow the pre-feedback slope to differ by stance
manip_pooled_r <- lm(meta_post ~ feedback * stance * moral + meta_pre_c * stance, data = s4)
show_tbl(anova_table(manip_pooled_r) |> filter(str_detect(term, "feedback")),
         "Robustness: stance-specific pre-feedback slopes")
```

Table 27: Robustness: stance-specific pre-feedback slopes

term	df1	df2	F	p	eta2_p	eta2_CI_low	eta2_CI_high
feedback	2	1190	730.214	0.000	0.551	0.523	1
feedback:stance	2	1190	105.824	0.000	0.151	0.121	1
feedback:moral	2	1190	5.857	0.003	0.010	0.002	1
feedback:stance:moral	2	1190	3.316	0.037	0.006	0.000	1

```r
s4 |>
  pivot_longer(c(meta_pre, meta_post), names_to = "time", values_to = "meta") |>
  mutate(time = factor(time, levels = c("meta_pre", "meta_post"), labels = c("Pre", "Post"))) |>
  group_by(stance, moral, feedback, time) |>
  summarise(M = mean(meta), SE = sd(meta) / sqrt(n()), .groups = "drop") |>
  ggplot(aes(time, M, color = feedback, group = feedback)) +
  geom_point(position = position_dodge(.2)) +
  geom_line(position = position_dodge(.2)) +
  geom_errorbar(aes(ymin = M - 1.96 * SE, ymax = M + 1.96 * SE), width = .1,
                position = position_dodge(.2)) +
  facet_grid(moral ~ stance) +
  labs(y = "Metaperception (1-5)", x = NULL, color = "Feedback",
       title = "Metaperception before and after feedback (means ± 95% CI)") +
  theme_minimal()
```

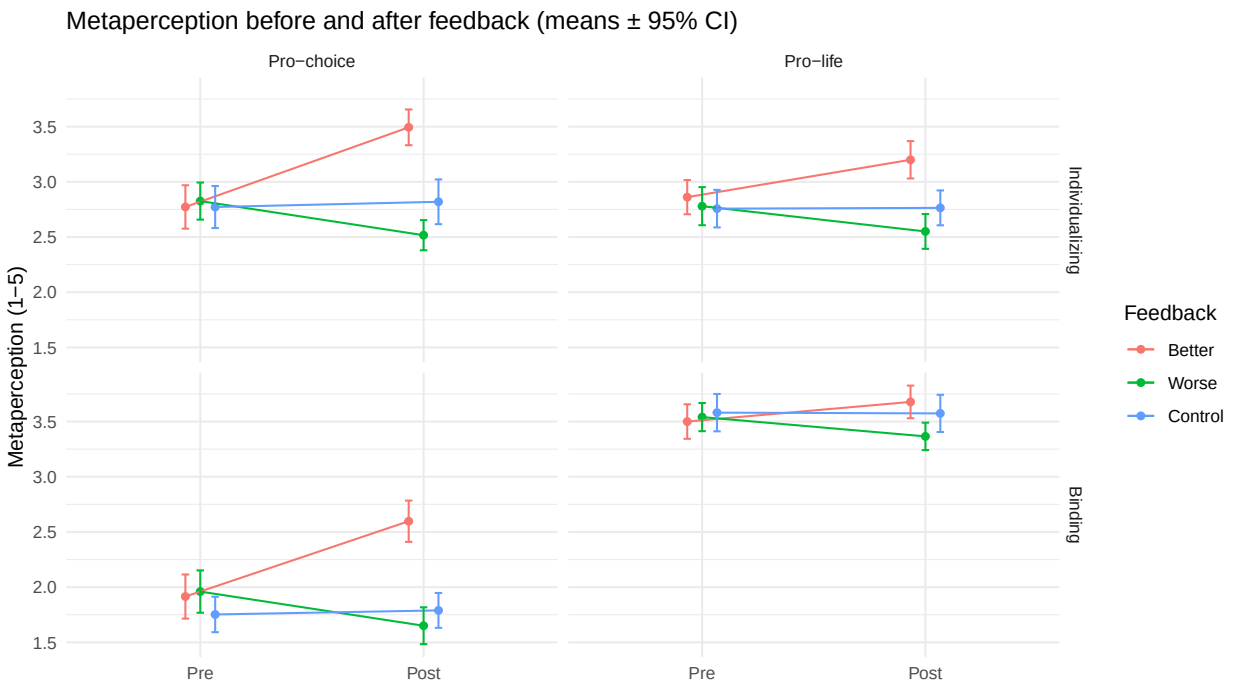


Figure 3

5. Preregistered confirmatory models

The preregistration specifies separate linear regressions for trust and threat on feedback, moral condition, perspective taking, and all their interactions. Ideology is "considered as potential covariate" and is therefore left out of the confirmatory model and added in Section 6. Perspective taking is the preregistered moderator; the empathy composite is secondary.

Preregistered hypotheses:

- **H1:** Better-than-expected → higher trust, lower threat.
- **H2:** Worse-than-expected → lower trust, higher threat.
- **Moderation:** higher perspective taking → stronger response to feedback.

Contrasts against Control are the direct tests of H1 and H2; Better vs. Worse is also reported.

```
dvs <- c("trust", "threat")

prereg_fits <- map(set_names(dvs), \(dv)
  map(by_stance, \(d) lm(reformulate("feedback * moral * PT_c", dv), data = d)))

prereg_anova <- imap_dfr(prereg_fits, \(fits, dv)
  imap_dfr(fits, \(f, s) anova_table(f) |> mutate(dv = dv, stance = s, .before = 1)))
show_tbl(prereg_anova, "Preregistered model: Type III tests")
```

Table 28: Preregistered model: Type III tests

dv	stance	term	df1	df2	F	p	eta2_p	eta2_CI_low	eta2_CI_high
trust	Pro-choice	feedback	2	592	47.321	0.000	0.138	0.097	1
trust	Pro-choice	moral	1	592	0.238	0.626	0.000	0.000	1
trust	Pro-choice	PT_c	1	592	23.564	0.000	0.038	0.017	1
trust	Pro-choice	feedback:moral	2	592	2.816	0.061	0.009	0.000	1
trust	Pro-choice	feedback:PT_c	2	592	0.608	0.545	0.002	0.000	1
trust	Pro-choice	moral:PT_c	1	592	0.002	0.964	0.000	0.000	1
trust	Pro-choice	feedback:moral:PT_c	2	592	0.101	0.904	0.000	0.000	1
trust	Pro-life	feedback	2	588	10.373	0.000	0.034	0.013	1
trust	Pro-life	moral	1	588	0.931	0.335	0.002	0.000	1
trust	Pro-life	PT_c	1	588	19.657	0.000	0.032	0.013	1
trust	Pro-life	feedback:moral	2	588	1.514	0.221	0.005	0.000	1
trust	Pro-life	feedback:PT_c	2	588	4.678	0.010	0.016	0.002	1
trust	Pro-life	moral:PT_c	1	588	5.308	0.022	0.009	0.001	1
trust	Pro-life	feedback:moral:PT_c	2	588	1.295	0.275	0.004	0.000	1
threat	Pro-choice	feedback	2	592	65.136	0.000	0.180	0.135	1
threat	Pro-choice	moral	1	592	0.040	0.841	0.000	0.000	1
threat	Pro-choice	PT_c	1	592	2.357	0.125	0.004	0.000	1
threat	Pro-choice	feedback:moral	2	592	1.923	0.147	0.006	0.000	1
threat	Pro-choice	feedback:PT_c	2	592	0.266	0.766	0.001	0.000	1
threat	Pro-choice	moral:PT_c	1	592	0.159	0.690	0.000	0.000	1
threat	Pro-choice	feedback:moral:PT_c	2	592	0.535	0.586	0.002	0.000	1
threat	Pro-life	feedback	2	588	9.418	0.000	0.031	0.011	1
threat	Pro-life	moral	1	588	2.107	0.147	0.004	0.000	1
threat	Pro-life	PT_c	1	588	5.822	0.016	0.010	0.001	1
threat	Pro-life	feedback:moral	2	588	1.888	0.152	0.006	0.000	1
threat	Pro-life	feedback:PT_c	2	588	5.356	0.005	0.018	0.003	1
threat	Pro-life	moral:PT_c	1	588	1.322	0.251	0.002	0.000	1
threat	Pro-life	feedback:moral:PT_c	2	588	3.767	0.024	0.013	0.001	1

```
# H1 and H2: each feedback condition vs. Control (Dunnett), within moral condition and collapsed
h_tests <- imap_dfr(prereg_fits, \(fits, dv) imap_dfr(fits, \(f, s) {
  within <- summary(contrast(emmeans(f, ~ feedback | moral), method = "trt.vs.ctrl", ref = "Control"),
                    infer = TRUE) |> as_tibble()
  collapsed <- summary(contrast(emmeans(f, ~ feedback), method = "trt.vs.ctrl", ref = "Control"),
                       infer = TRUE) |> as_tibble() |> mutate(moral = "Averaged")
  bind_rows(within |> mutate(moral = as.character(moral)), collapsed) |>
    mutate(dv = dv, stance = s, .before = 1)
}))
show_tbl(h_tests, "H1 (Better - Control) and H2 (Worse - Control), at mean perspective taking")
```

Table 29: H1 (Better - Control) and H2 (Worse - Control), at mean perspective taking

dv	stance	contrast	moral	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
trust	Pro-choice	Better - Control	Individualizing	1.259	0.184	592	0.850	1.667	6.855	0.000
trust	Pro-choice	Worse - Control	Individualizing	0.005	0.181	592	-0.398	0.409	0.030	0.998
trust	Pro-choice	Better - Control	Binding	0.658	0.187	592	0.241	1.075	3.516	0.001
trust	Pro-choice	Worse - Control	Binding	-0.439	0.184	592	-0.849	-0.029	-2.382	0.034
trust	Pro-choice	Better - Control	Averaged	0.958	0.131	592	0.666	1.250	7.310	0.000
trust	Pro-choice	Worse - Control	Averaged	-0.217	0.129	592	-0.504	0.071	-1.679	0.169
trust	Pro-life	Better - Control	Individualizing	0.472	0.195	588	0.038	0.906	2.425	0.030
trust	Pro-life	Worse - Control	Individualizing	-0.257	0.195	588	-0.691	0.177	-1.320	0.320
trust	Pro-life	Better - Control	Binding	-0.003	0.192	588	-0.431	0.424	-0.018	0.999
trust	Pro-life	Worse - Control	Binding	-0.489	0.192	588	-0.917	-0.061	-2.543	0.022
trust	Pro-life	Better - Control	Averaged	0.234	0.137	588	-0.070	0.539	1.714	0.158
trust	Pro-life	Worse - Control	Averaged	-0.373	0.137	588	-0.678	-0.068	-2.726	0.013
threat	Pro-choice	Better - Control	Individualizing	-1.494	0.204	592	-1.949	-1.038	-7.306	0.000
threat	Pro-choice	Worse - Control	Individualizing	0.260	0.202	592	-0.189	0.709	1.289	0.336
threat	Pro-choice	Better - Control	Binding	-0.946	0.208	592	-1.411	-0.482	-4.541	0.000
threat	Pro-choice	Worse - Control	Binding	0.399	0.205	592	-0.058	0.856	1.944	0.097
threat	Pro-choice	Better - Control	Averaged	-1.220	0.146	592	-1.545	-0.895	-8.358	0.000
threat	Pro-choice	Worse - Control	Averaged	0.329	0.144	592	0.009	0.650	2.291	0.043
threat	Pro-life	Better - Control	Individualizing	-0.541	0.203	588	-0.993	-0.089	-2.663	0.015
threat	Pro-life	Worse - Control	Individualizing	0.232	0.203	588	-0.220	0.685	1.143	0.417
threat	Pro-life	Better - Control	Binding	0.008	0.200	588	-0.438	0.454	0.039	0.997
threat	Pro-life	Worse - Control	Binding	0.449	0.201	588	0.002	0.896	2.239	0.049
threat	Pro-life	Better - Control	Averaged	-0.267	0.143	588	-0.584	0.051	-1.868	0.115
threat	Pro-life	Worse - Control	Averaged	0.341	0.143	588	0.023	0.659	2.387	0.033

```
all_pairs <- imap_dfr(prereg_fits, \(fits, dv) imap_dfr(fits, \(f, s)
  pairs_table(f, ~ feedback | moral) |> mutate(dv = dv, stance = s, .before = 1)))
show_tbl(all_pairs, "All pairwise feedback contrasts within moral condition (Tukey), at mean PT")
```

Table 30: All pairwise feedback contrasts within moral condition (Tukey), at mean PT

dv	stance	contrast	moral	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
trust	Pro-choice	Better - Worse	Individualizing	1.253	0.177	592	0.837	1.669	7.083	0.000
trust	Pro-choice	Better - Control	Individualizing	1.259	0.184	592	0.827	1.690	6.855	0.000
trust	Pro-choice	Worse - Control	Individualizing	0.005	0.181	592	-0.420	0.431	0.030	0.999
trust	Pro-choice	Better - Worse	Binding	1.097	0.184	592	0.666	1.528	5.977	0.000
trust	Pro-choice	Better - Control	Binding	0.658	0.187	592	0.218	1.098	3.516	0.001
trust	Pro-choice	Worse - Control	Binding	-0.439	0.184	592	-0.872	-0.006	-2.382	0.046
trust	Pro-life	Better - Worse	Individualizing	0.729	0.191	588	0.280	1.178	3.817	0.000
trust	Pro-life	Better - Control	Individualizing	0.472	0.195	588	0.015	0.929	2.425	0.041
trust	Pro-life	Worse - Control	Individualizing	-0.257	0.195	588	-0.715	0.201	-1.320	0.385
trust	Pro-life	Better - Worse	Binding	0.486	0.189	588	0.041	0.930	2.566	0.028
trust	Pro-life	Better - Control	Binding	-0.003	0.192	588	-0.455	0.448	-0.018	1.000
trust	Pro-life	Worse - Control	Binding	-0.489	0.192	588	-0.941	-0.037	-2.543	0.030
threat	Pro-choice	Better - Worse	Individualizing	-1.754	0.197	592	-2.217	-1.291	-8.900	0.000
threat	Pro-choice	Better - Control	Individualizing	-1.494	0.204	592	-1.974	-1.013	-7.306	0.000
threat	Pro-choice	Worse - Control	Individualizing	0.260	0.202	592	-0.214	0.733	1.289	0.402
threat	Pro-choice	Better - Worse	Binding	-1.345	0.204	592	-1.826	-0.865	-6.582	0.000
threat	Pro-choice	Better - Control	Binding	-0.946	0.208	592	-1.436	-0.457	-4.541	0.000
threat	Pro-choice	Worse - Control	Binding	0.399	0.205	592	-0.083	0.881	1.944	0.127
threat	Pro-life	Better - Worse	Individualizing	-0.773	0.199	588	-1.242	-0.305	-3.880	0.000
threat	Pro-life	Better - Control	Individualizing	-0.541	0.203	588	-1.018	-0.064	-2.663	0.022
threat	Pro-life	Worse - Control	Individualizing	0.232	0.203	588	-0.245	0.710	1.143	0.488
threat	Pro-life	Better - Worse	Binding	-0.441	0.197	588	-0.905	0.022	-2.236	0.066
threat	Pro-life	Better - Control	Binding	0.008	0.200	588	-0.463	0.479	0.039	0.999
threat	Pro-life	Worse - Control	Binding	0.449	0.201	588	-0.022	0.921	2.239	0.066

```
# Perspective-taking slopes by feedback condition and their differences
pt_slopes <- imap_dfr(prereg_fits, \(fits, dv) imap_dfr(fits, \(f, s)
  summary(emtrends(f, ~ feedback, var = "PT_c"), infer = TRUE) |> as_tibble() |>
```

```
    mutate(dv = dv, stance = s, .before = 1)))
show_tbl(pt_slopes, "Perspective-taking slope within each feedback condition")
```

Table 31: Perspective-taking slope within each feedback condition

dv	stance	feedback	PT_c.trend	SE	df	lower.CL	upper.CL	t.ratio	p.value
trust	Pro-choice	Better	0.249	0.126	592	0.001	0.496	1.975	0.049
trust	Pro-choice	Worse	0.440	0.135	592	0.176	0.705	3.269	0.001
trust	Pro-choice	Control	0.396	0.126	592	0.148	0.644	3.135	0.002
trust	Pro-life	Better	0.649	0.123	588	0.407	0.890	5.279	0.000
trust	Pro-life	Worse	0.133	0.136	588	-0.134	0.400	0.981	0.327
trust	Pro-life	Control	0.223	0.133	588	-0.039	0.484	1.670	0.095
threat	Pro-choice	Better	-0.138	0.140	592	-0.414	0.137	-0.986	0.325
threat	Pro-choice	Worse	-0.196	0.150	592	-0.490	0.099	-1.306	0.192
threat	Pro-choice	Control	-0.048	0.141	592	-0.324	0.228	-0.341	0.733
threat	Pro-life	Better	-0.524	0.128	588	-0.776	-0.272	-4.089	0.000
threat	Pro-life	Worse	0.088	0.142	588	-0.191	0.366	0.618	0.537
threat	Pro-life	Control	-0.134	0.139	588	-0.407	0.139	-0.963	0.336

```
pt_slope_diffs <- imap_dfr(prereg_fits, \(fits, dv) imap_dfr(fits, \(f, s)
  summary(pairs(emtrends(f, ~ feedback, var = "PT_c")), infer = TRUE) |> as_tibble() |>
    mutate(dv = dv, stance = s, .before = 1)))
show_tbl(pt_slope_diffs, "Differences in PT slopes between feedback conditions (Tukey)")
```

Table 32: Differences in PT slopes between feedback conditions (Tukey)

dv	stance	contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
trust	Pro-choice	Better - Worse	-0.191	0.184	592	-0.625	0.242	-1.039	0.553
trust	Pro-choice	Better - Control	-0.147	0.178	592	-0.566	0.272	-0.825	0.688
trust	Pro-choice	Worse - Control	0.044	0.185	592	-0.389	0.478	0.241	0.969
trust	Pro-life	Better - Worse	0.515	0.183	588	0.085	0.946	2.812	0.014
trust	Pro-life	Better - Control	0.426	0.181	588	0.000	0.852	2.350	0.050
trust	Pro-life	Worse - Control	-0.089	0.190	588	-0.536	0.358	-0.468	0.886
threat	Pro-choice	Better - Worse	0.058	0.205	592	-0.425	0.540	0.280	0.958
threat	Pro-choice	Better - Control	-0.090	0.199	592	-0.557	0.376	-0.455	0.892
threat	Pro-choice	Worse - Control	-0.148	0.206	592	-0.631	0.335	-0.719	0.752
threat	Pro-life	Better - Worse	-0.612	0.191	588	-1.061	-0.163	-3.201	0.004
threat	Pro-life	Better - Control	-0.390	0.189	588	-0.835	0.054	-2.064	0.098
threat	Pro-life	Worse - Control	0.222	0.199	588	-0.245	0.688	1.115	0.505

```
# One-line verdicts for the write-up (averaged over moral, at mean PT)
h_tests |>
  filter(moral == "Averaged") |>
  mutate(
    hypothesis = if_else(str_detect(contrast, "Better"), "H1", "H2"),
    predicted_sign = case_when(
      hypothesis == "H1" & dv == "trust"  ~  1,
      hypothesis == "H1" & dv == "threat" ~ -1,
      hypothesis == "H2" & dv == "trust"  ~ -1,
      hypothesis == "H2" & dv == "threat" ~  1),
    supported = p.value < .05 & sign(estimate) == predicted_sign
  ) |>
  select(stance, dv, hypothesis, contrast, estimate, p.value, supported) |>
  show_tbl("Preregistered hypothesis verdicts (vs. Control, averaged over moral condition)")
```

Table 33: Preregistered hypothesis verdicts (vs. Control, averaged over moral condition)

stance	dv	hypothesis	contrast	estimate	p.value	supported
Pro-choice	trust	H1	Better - Control	0.958	0.000	TRUE
Pro-choice	trust	H2	Worse - Control	-0.217	0.169	FALSE
Pro-life	trust	H1	Better - Control	0.234	0.158	FALSE
Pro-life	trust	H2	Worse - Control	-0.373	0.013	TRUE
Pro-choice	threat	H1	Better - Control	-1.220	0.000	TRUE
Pro-choice	threat	H2	Worse - Control	0.329	0.043	TRUE
Pro-life	threat	H1	Better - Control	-0.267	0.115	FALSE
Pro-life	threat	H2	Worse - Control	0.341	0.033	TRUE

6. Robustness

Robustness specifications:

- **R1:** the ANCOVA reported in the submitted manuscript (ideology + feedback × moral). The F values should match the SI. If they don't, the discrepancy traces to the stored-score problem flagged in Section 1.
- **R2:** R1 plus pre-feedback metaperception, which adds precision.
- **R3:** the preregistered model with the empathy composite (EC + PT) in place of PT.

```
specs <- list(
  R1_original_ANCOVA = "feedback * moral + ideology",
  R2_plus_meta_pre   = "feedback * moral + ideology + meta_pre_c",
  R3_empathy_comp    = "feedback * moral * empathy_c"
)

robust <- map_dfr(names(specs), \(spec) map_dfr(dvs, \(dv) imap_dfr(by_stance, \(d, s) {
  f <- lm(reformulate(specs[[spec]], dv), data = d)
  a <- anova_table(f) |> filter(term %in% c("feedback", "moral", "feedback:moral"))
  pr <- pairs_table(f, ~ feedback) |> select(contrast, estimate, SE, p.value)
  a |>
    mutate(spec = spec, dv = dv, stance = s, .before = 1) |>
    mutate(BW_est = pr$estimate[pr$contrast == "Better - Worse"],
           BW_p   = pr$p.value[pr$contrast == "Better - Worse"],
           BC_est = pr$estimate[pr$contrast == "Better - Control"],
           BC_p   = pr$p.value[pr$contrast == "Better - Control"])
})))
show_tbl(robust, "Robustness: feedback effects across specifications (B = Better, W = Worse, C = Control)")
```

Table 34: Robustness: feedback effects across specifications (B = Better, W = Worse, C = Control)

spec	dv	stance	term	df1	df2	F	p	eta2_p	eta2_CI_low	eta2_CI_high	BW_est	BW_p	BC_est	BC_p
R1_original_ANCOVA	trust	Pro-choice	feedback	2	597	45.668	0.000	0.133	0.092	1	1.126	0	0.880	0.000
R1_original_ANCOVA	trust	Pro-choice	moral	1	597	0.706	0.401	0.001	0.000	1	1.126	0	0.880	0.000
R1_original_ANCOVA	trust	Pro-choice	feedback:moral	2	597	1.573	0.208	0.005	0.000	1	1.126	0	0.880	0.000
R1_original_ANCOVA	trust	Pro-life	feedback	2	593	9.438	0.000	0.031	0.011	1	0.587	0	0.249	0.170
R1_original_ANCOVA	trust	Pro-life	moral	1	593	0.785	0.376	0.001	0.000	1	0.587	0	0.249	0.170
R1_original_ANCOVA	trust	Pro-life	feedback:moral	2	593	0.872	0.419	0.003	0.000	1	0.587	0	0.249	0.170
R1_original_ANCOVA	threat	Pro-choice	feedback	2	597	68.315	0.000	0.186	0.141	1	-1.498	0	-1.121	0.000
R1_original_ANCOVA	threat	Pro-choice	moral	1	597	0.325	0.569	0.001	0.000	1	-1.498	0	-1.121	0.000
R1_original_ANCOVA	threat	Pro-choice	feedback:moral	2	597	1.230	0.293	0.004	0.000	1	-1.498	0	-1.121	0.000
R1_original_ANCOVA	threat	Pro-life	feedback	2	593	8.531	0.000	0.028	0.009	1	-0.572	0	-0.281	0.114
R1_original_ANCOVA	threat	Pro-life	moral	1	593	1.577	0.210	0.003	0.000	1	-0.572	0	-0.281	0.114
R1_original_ANCOVA	threat	Pro-life	feedback:moral	2	593	1.259	0.285	0.004	0.000	1	-0.572	0	-0.281	0.114
R2_plus_meta_pre	trust	Pro-choice	feedback	2	596	48.589	0.000	0.140	0.099	1	1.143	0	0.864	0.000
R2_plus_meta_pre	trust	Pro-choice	moral	1	596	9.959	0.002	0.016	0.004	1	1.143	0	0.864	0.000
R2_plus_meta_pre	trust	Pro-choice	feedback:moral	2	596	2.124	0.120	0.007	0.000	1	1.143	0	0.864	0.000
R2_plus_meta_pre	trust	Pro-life	feedback	2	592	9.438	0.000	0.031	0.011	1	0.579	0	0.246	0.169
R2_plus_meta_pre	trust	Pro-life	moral	1	592	6.464	0.011	0.011	0.001	1	0.579	0	0.246	0.169
R2_plus_meta_pre	trust	Pro-life	feedback:moral	2	592	0.637	0.529	0.002	0.000	1	0.579	0	0.246	0.169
R2_plus_meta_pre	threat	Pro-choice	feedback	2	596	71.344	0.000	0.193	0.147	1	-1.513	0	-1.107	0.000
R2_plus_meta_pre	threat	Pro-choice	moral	1	596	6.240	0.013	0.010	0.001	1	-1.513	0	-1.107	0.000
R2_plus_meta_pre	threat	Pro-choice	feedback:moral	2	596	1.508	0.222	0.005	0.000	1	-1.513	0	-1.107	0.000
R2_plus_meta_pre	threat	Pro-life	feedback	2	592	8.558	0.000	0.028	0.009	1	-0.574	0	-0.282	0.114
R2_plus_meta_pre	threat	Pro-life	moral	1	592	0.762	0.383	0.001	0.000	1	-0.574	0	-0.282	0.114
R2_plus_meta_pre	threat	Pro-life	feedback:moral	2	592	1.307	0.272	0.004	0.000	1	-0.574	0	-0.282	0.114
R3_empathy_comp	trust	Pro-choice	feedback	2	592	45.979	0.000	0.134	0.094	1	1.167	0	0.970	0.000
R3_empathy_comp	trust	Pro-choice	moral	1	592	0.316	0.574	0.001	0.000	1	1.167	0	0.970	0.000
R3_empathy_comp	trust	Pro-choice	feedback:moral	2	592	2.535	0.080	0.008	0.000	1	1.167	0	0.970	0.000
R3_empathy_comp	trust	Pro-life	feedback	2	588	9.921	0.000	0.033	0.012	1	0.601	0	0.242	0.186
R3_empathy_comp	trust	Pro-life	moral	1	588	0.768	0.381	0.001	0.000	1	0.601	0	0.242	0.186
R3_empathy_comp	trust	Pro-life	feedback:moral	2	588	1.157	0.315	0.004	0.000	1	0.601	0	0.242	0.186
R3_empathy_comp	threat	Pro-choice	feedback	2	592	64.630	0.000	0.179	0.134	1	-1.543	0	-1.227	0.000
R3_empathy_comp	threat	Pro-choice	moral	1	592	0.050	0.823	0.000	0.000	1	-1.543	0	-1.227	0.000
R3_empathy_comp	threat	Pro-choice	feedback:moral	2	592	1.853	0.158	0.006	0.000	1	-1.543	0	-1.227	0.000
R3_empathy_comp	threat	Pro-life	feedback	2	588	8.813	0.000	0.029	0.010	1	-0.589	0	-0.260	0.164
R3_empathy_comp	threat	Pro-life	moral	1	588	2.142	0.144	0.004	0.000	1	-0.589	0	-0.260	0.164
R3_empathy_comp	threat	Pro-life	feedback:moral	2	588	1.555	0.212	0.005	0.000	1	-0.589	0	-0.260	0.164

```
# Re-estimate the empathy moderation reported in the submitted manuscript, now with reverse scoring
emp_fits <- map(set_names(dvs), \(dv)
  map(by_stance, \(d) lm(reformulate("feedback * moral * empathy_c + ideology", dv), data = d)))

imap_dfr(emp_fits, \(fits, dv) imap_dfr(fits, \(f, s)
  anova_table(f) |> filter(str_detect(term, "empathy")) |> mutate(dv = dv, stance = s, .before = 1))) |>
  show_tbl("Empathy composite moderation (with ideology), Type III")
```

Table 35: Empathy composite moderation (with ideology), Type III

dv	stance	term	df1	df2	F	p	eta2_p	eta2_CI_low	eta2_CI_high
trust	Pro-choice	empathy_c	1	591	13.273	0.000	0.022	0.007	1
trust	Pro-choice	feedback:empathy_c	2	591	0.439	0.645	0.001	0.000	1
trust	Pro-choice	moral:empathy_c	1	591	0.512	0.475	0.001	0.000	1
trust	Pro-choice	feedback:moral:empathy_c	2	591	0.715	0.489	0.002	0.000	1
trust	Pro-life	empathy_c	1	587	10.030	0.002	0.017	0.004	1
trust	Pro-life	feedback:empathy_c	2	587	3.890	0.021	0.013	0.001	1
trust	Pro-life	moral:empathy_c	1	587	5.871	0.016	0.010	0.001	1
trust	Pro-life	feedback:moral:empathy_c	2	587	1.701	0.183	0.006	0.000	1
threat	Pro-choice	empathy_c	1	591	0.125	0.724	0.000	0.000	1
threat	Pro-choice	feedback:empathy_c	2	591	0.286	0.751	0.001	0.000	1
threat	Pro-choice	moral:empathy_c	1	591	0.012	0.914	0.000	0.000	1
threat	Pro-choice	feedback:moral:empathy_c	2	591	0.082	0.921	0.000	0.000	1
threat	Pro-life	empathy_c	1	587	1.269	0.260	0.002	0.000	1
threat	Pro-life	feedback:empathy_c	2	587	6.541	0.002	0.022	0.005	1
threat	Pro-life	moral:empathy_c	1	587	4.615	0.032	0.008	0.000	1
threat	Pro-life	feedback:moral:empathy_c	2	587	5.323	0.005	0.018	0.003	1

```
imap_dfr(emp_fits, \(fits, dv) imap_dfr(fits, \(f, s)
  summary(emtrends(f, ~ feedback, var = "empathy_c"), infer = TRUE) |> as_tibble() |>
    mutate(dv = dv, stance = s, .before = 1))) |>
  show_tbl("Empathy composite slopes by feedback condition")
```

Table 36: Empathy composite slopes by feedback condition

dv	stance	feedback	empathy_c.trend	SE	df	lower.CL	upper.CL	t.ratio	p.value
trust	Pro-choice	Better	0.280	0.129	591	0.027	0.533	2.175	0.030
trust	Pro-choice	Worse	0.202	0.140	591	-0.073	0.478	1.442	0.150
trust	Pro-choice	Control	0.390	0.144	591	0.107	0.673	2.706	0.007
trust	Pro-life	Better	0.599	0.137	587	0.329	0.869	4.362	0.000
trust	Pro-life	Worse	0.074	0.156	587	-0.232	0.380	0.474	0.636
trust	Pro-life	Control	0.148	0.155	587	-0.156	0.453	0.958	0.339
threat	Pro-choice	Better	-0.117	0.141	591	-0.393	0.158	-0.836	0.403
threat	Pro-choice	Worse	-0.010	0.153	591	-0.310	0.291	-0.063	0.950
threat	Pro-choice	Control	0.035	0.157	591	-0.274	0.344	0.222	0.824
threat	Pro-life	Better	-0.499	0.141	587	-0.775	-0.223	-3.551	0.000
threat	Pro-life	Worse	0.262	0.160	587	-0.052	0.576	1.641	0.101
threat	Pro-life	Control	-0.062	0.159	587	-0.374	0.250	-0.391	0.696

7. Pooled stance × feedback models (Comment 3)

This is a single model on the full sample. The key term is **feedback × stance**. Ideology is centered within stance so it doesn't absorb the stance effect. Results are reported both without ideology (P1) and with it (P2).

```
pooled_specs <- list(
  P1_no_covariate = "feedback * stance * moral",
  P2_ideo_within  = "feedback * stance * moral + ideo_c"
)

pooled_fits <- map(set_names(names(pooled_specs)), \(spec)
  map(set_names(dvs), \(dv) lm(reformulate(pooled_specs[[spec]], dv), data = s4)))

pooled_anova <- imap_dfr(pooled_fits, \(fits, spec) imap_dfr(fits, \(f, dv)
  anova_table(f) |> mutate(spec = spec, dv = dv, .before = 1)))
show_tbl(pooled_anova, "Pooled models: Type III tests (focal term: feedback:stance)")
```

Table 37: Pooled models: Type III tests (focal term: feedback:stance)

spec	dv	term	df1	df2	F	p	eta2_p	eta2_CI_low	eta2_CI_high
P1_no_covariate	trust	feedback	2	1192	45.771	0.000	0.071	0.049	1
P1_no_covariate	trust	stance	1	1192	1.804	0.180	0.002	0.000	1
P1_no_covariate	trust	moral	1	1192	0.100	0.752	0.000	0.000	1
P1_no_covariate	trust	feedback:stance	2	1192	7.618	0.001	0.013	0.004	1
P1_no_covariate	trust	feedback:moral	2	1192	3.214	0.041	0.005	0.000	1
P1_no_covariate	trust	stance:moral	1	1192	1.200	0.274	0.001	0.000	1
P1_no_covariate	trust	feedback:stance:moral	2	1192	0.082	0.921	0.000	0.000	1
P1_no_covariate	threat	feedback	2	1192	59.913	0.000	0.091	0.066	1
P1_no_covariate	threat	stance	1	1192	23.556	0.000	0.019	0.009	1
P1_no_covariate	threat	moral	1	1192	0.578	0.447	0.000	0.000	1
P1_no_covariate	threat	feedback:stance	2	1192	14.309	0.000	0.023	0.011	1
P1_no_covariate	threat	feedback:moral	2	1192	3.401	0.034	0.006	0.000	1
P1_no_covariate	threat	stance:moral	1	1192	1.308	0.253	0.001	0.000	1
P1_no_covariate	threat	feedback:stance:moral	2	1192	0.034	0.967	0.000	0.000	1
P2_ideo_within	trust	feedback	2	1191	45.820	0.000	0.071	0.049	1
P2_ideo_within	trust	stance	1	1191	1.820	0.178	0.002	0.000	1
P2_ideo_within	trust	moral	1	1191	0.091	0.762	0.000	0.000	1
P2_ideo_within	trust	ideo_c	1	1191	6.058	0.014	0.005	0.001	1
P2_ideo_within	trust	feedback:stance	2	1191	7.228	0.001	0.012	0.003	1
P2_ideo_within	trust	feedback:moral	2	1191	3.182	0.042	0.005	0.000	1
P2_ideo_within	trust	stance:moral	1	1191	1.349	0.246	0.001	0.000	1
P2_ideo_within	trust	feedback:stance:moral	2	1191	0.044	0.957	0.000	0.000	1
P2_ideo_within	threat	feedback	2	1191	60.027	0.000	0.092	0.067	1
P2_ideo_within	threat	stance	1	1191	23.724	0.000	0.020	0.009	1
P2_ideo_within	threat	moral	1	1191	0.556	0.456	0.000	0.000	1
P2_ideo_within	threat	ideo_c	1	1191	7.677	0.006	0.006	0.001	1
P2_ideo_within	threat	feedback:stance	2	1191	13.711	0.000	0.023	0.010	1
P2_ideo_within	threat	feedback:moral	2	1191	3.342	0.036	0.006	0.000	1
P2_ideo_within	threat	stance:moral	1	1191	1.485	0.223	0.001	0.000	1
P2_ideo_within	threat	feedback:stance:moral	2	1191	0.044	0.957	0.000	0.000	1

```
pooled_int <- imap_dfr(pooled_fits, \(fits, spec) imap_dfr(fits, \(f, dv)
  summary(contrast(emmeans(f, ~ feedback * stance),
                   interaction = c("pairwise", "pairwise"), adjust = "holm"),
          infer = TRUE) |> as_tibble() |>
    mutate(spec = spec, dv = dv, .before = 1)))
show_tbl(pooled_int,
         "Interaction contrasts: (feedback contrast among pro-choice) - (same contrast among pro-life)")
```

Table 38: Interaction contrasts: (feedback contrast among pro-choice) - (same contrast among pro-life)

spec	dv	feedback_pairwise	stance_pairwise	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
P1_no_covariate	trust	Better - Worse	(Pro-choice) - (Pro-life)	0.542	0.189	1192	0.090	0.995	2.874	0.008
P1_no_covariate	trust	Better - Control	(Pro-choice) - (Pro-life)	0.719	0.193	1192	0.256	1.182	3.721	0.001
P1_no_covariate	trust	Worse - Control	(Pro-choice) - (Pro-life)	0.177	0.192	1192	-0.282	0.636	0.923	0.356
P1_no_covariate	threat	Better - Worse	(Pro-choice) - (Pro-life)	-0.929	0.200	1192	-1.409	-0.448	-4.633	0.000
P1_no_covariate	threat	Better - Control	(Pro-choice) - (Pro-life)	-0.949	0.205	1192	-1.441	-0.457	-4.625	0.000
P1_no_covariate	threat	Worse - Control	(Pro-choice) - (Pro-life)	-0.021	0.203	1192	-0.508	0.467	-0.101	0.919
P2_ideo_within	trust	Better - Worse	(Pro-choice) - (Pro-life)	0.524	0.188	1191	0.073	0.976	2.782	0.011
P2_ideo_within	trust	Better - Control	(Pro-choice) - (Pro-life)	0.701	0.193	1191	0.238	1.163	3.633	0.001
P2_ideo_within	trust	Worse - Control	(Pro-choice) - (Pro-life)	0.177	0.191	1191	-0.282	0.635	0.924	0.356
P2_ideo_within	threat	Better - Worse	(Pro-choice) - (Pro-life)	-0.907	0.200	1191	-1.387	-0.428	-4.535	0.000
P2_ideo_within	threat	Better - Control	(Pro-choice) - (Pro-life)	-0.928	0.205	1191	-1.419	-0.437	-4.529	0.000
P2_ideo_within	threat	Worse - Control	(Pro-choice) - (Pro-life)	-0.021	0.203	1191	-0.507	0.466	-0.101	0.919

```
pooled_simple <- imap_dfr(pooled_fits$P1_no_covariate, \(f, dv)
  pairs_table(f, ~ feedback | stance) |> mutate(dv = dv, .before = 1))
show_tbl(pooled_simple, "Simple feedback contrasts within stance (pooled P1 model, Tukey)")
```

Table 39: Simple feedback contrasts within stance (pooled P1 model, Tukey)

dv	contrast	stance	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
trust	Better - Worse	Pro-choice	1.158	0.133	1192	0.846	1.470	8.699	0.000
trust	Better - Control	Pro-choice	0.961	0.137	1192	0.639	1.283	7.012	0.000
trust	Worse - Control	Pro-choice	-0.197	0.135	1192	-0.513	0.120	-1.459	0.311
trust	Better - Worse	Pro-life	0.616	0.134	1192	0.302	0.929	4.602	0.000
trust	Better - Control	Pro-life	0.242	0.136	1192	-0.077	0.562	1.778	0.177
trust	Worse - Control	Pro-life	-0.373	0.136	1192	-0.693	-0.054	-2.745	0.017
threat	Better - Worse	Pro-choice	-1.538	0.141	1192	-1.870	-1.206	-10.876	0.000
threat	Better - Control	Pro-choice	-1.222	0.146	1192	-1.564	-0.881	-8.394	0.000
threat	Worse - Control	Pro-choice	0.316	0.143	1192	-0.021	0.652	2.203	0.071
threat	Better - Worse	Pro-life	-0.609	0.142	1192	-0.943	-0.276	-4.288	0.000
threat	Better - Control	Pro-life	-0.273	0.145	1192	-0.613	0.066	-1.887	0.143
threat	Worse - Control	Pro-life	0.336	0.145	1192	-0.003	0.675	2.327	0.053

```
s4 |>
  filter(feedback == "Control") |>
  group_by(stance) |>
  summarise(n = n(),
            trust_M = mean(trust), trust_SD = sd(trust),
            threat_M = mean(threat), threat_SD = sd(threat),
            meta_pre_M = mean(meta_pre)) |>
  show_tbl("Control-condition means by stance (promised in the Comment 3 response)", digits = 2)
```

Table 40: Control-condition means by stance (promised in the Comment 3 response)

stance	n	trust_M	trust_SD	threat_M	threat_SD	meta_pre_M
Pro-choice	190	4.17	1.50	4.01	1.66	2.26
Pro-life	191	4.37	1.31	4.09	1.41	3.18

```
s4 |>
  pivot_longer(c(trust, threat), names_to = "outcome", values_to = "score") |>
  ggplot(aes(feedback, score, fill = feedback)) +
  geom_violin(alpha = .4, color = NA) +
  stat_summary(fun.data = mean_cl_normal, geom = "pointrange") +
  facet_grid(outcome ~ stance) +
  labs(x = NULL, y = "Score (1-7)", title = "Trust and threat by feedback and stance (mean ± 95% CI)") +
  theme_minimal() + theme(legend.position = "none")
```

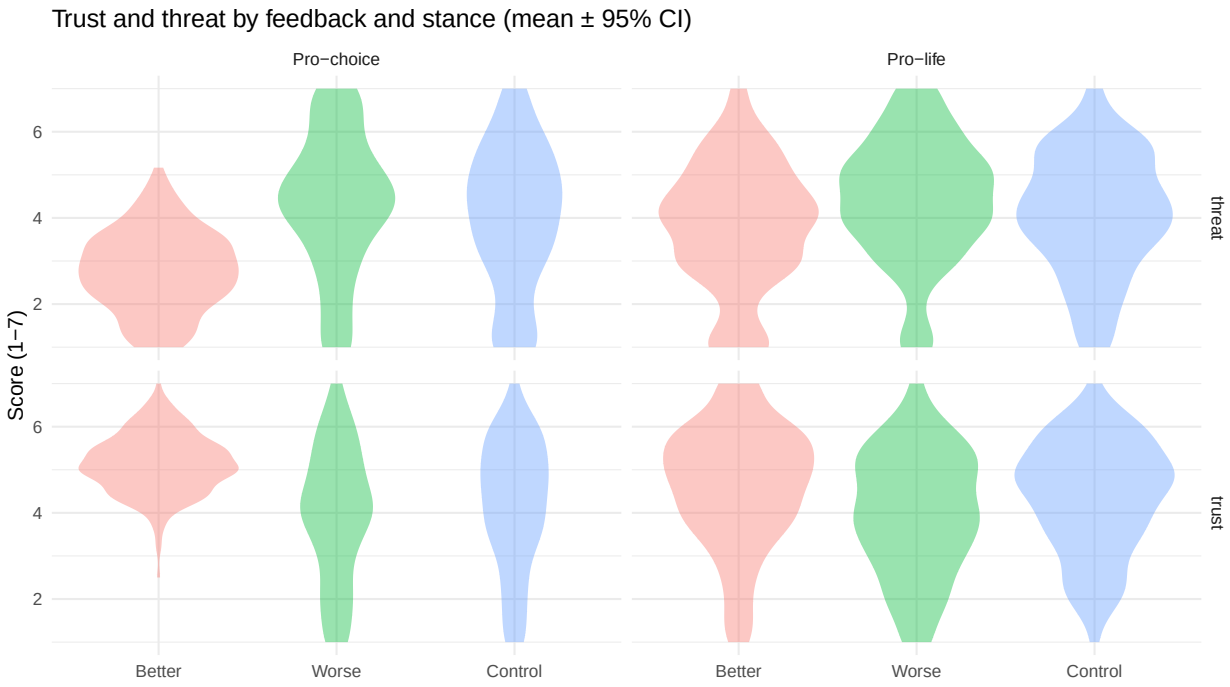


Figure 4

8. Exploratory: baseline-estimate analyses

Interpretive limits.

- A *linear* discrepancy account and a valence account both predict **no** feedback × baseline interaction. Because the displayed values are fixed, the between-condition difference in discrepancy is the same for every participant.
- Participants saw the group estimate (2.5), not their own.
- These analyses therefore cannot settle mechanism. Report them in the SI as exploratory.

8.1 Linear interaction (as promised in the response letter)

```
lin_fits <- map(set_names(dvs), \(dv)
  map(by_stance, \(d) lm(reformulate("feedback * meta_pre_c + moral + ideology", dv), data = d)))

imap_dfr(lin_fits, \(fits, dv) imap_dfr(fits, \(f, s)
  anova_table(f) |> filter(term %in% c("feedback", "meta_pre_c", "feedback:meta_pre_c")) |>
    mutate(dv = dv, stance = s, .before = 1))) |>
  show_tbl("Feedback x pre-feedback metaperception")
```

Table 41: Feedback x pre-feedback metaperception

dv	stance	term	df1	df2	F	p	eta2_p	eta2_CI_low	eta2_CI_high
trust	Pro-choice	feedback	2	596	49.453	0.000	0.142	0.101	1
trust	Pro-choice	meta_pre_c	1	596	30.936	0.000	0.049	0.025	1
trust	Pro-choice	feedback:meta_pre_c	2	596	3.339	0.036	0.011	0.000	1
trust	Pro-life	feedback	2	592	9.389	0.000	0.031	0.011	1
trust	Pro-life	meta_pre_c	1	592	17.622	0.000	0.029	0.011	1
trust	Pro-life	feedback:meta_pre_c	2	592	0.195	0.823	0.001	0.000	1
threat	Pro-choice	feedback	2	596	72.081	0.000	0.195	0.149	1
threat	Pro-choice	meta_pre_c	1	596	21.226	0.000	0.034	0.014	1
threat	Pro-choice	feedback:meta_pre_c	2	596	1.356	0.258	0.005	0.000	1
threat	Pro-life	feedback	2	592	8.546	0.000	0.028	0.009	1
threat	Pro-life	meta_pre_c	1	592	0.359	0.549	0.001	0.000	1
threat	Pro-life	feedback:meta_pre_c	2	592	1.510	0.222	0.005	0.000	1

8.2 Sign-based (kink) test

The kink term is `above_shown = max(0, pre - value shown)`. Its slope tests whether the relation between baseline estimate and outcome bends at **the value that condition displayed** (3.8, 2.8, or 1.2). The kink location differs by condition, so a common nonlinear baseline–outcome relation (absorbed by a quadratic) does not produce it. A valence account predicts a slope near 0.

```
s4 <- s4 |>
  mutate(value_shown = c(Better = 3.8, Worse = 1.2, Control = 2.8)[as.character(feedback)],
         above_shown = pmax(0, meta_pre - value_shown),
         pre_above_shown = meta_pre > value_shown)
by_stance <- split(s4, s4$stance)

s4 |> count(stance, feedback, pre_above_shown) |>
  pivot_wider(names_from = pre_above_shown, values_from = n, names_prefix = "pre_above_shown_") |>
  show_tbl("How many participants' own estimate exceeded the value shown")
```

Table 42: How many participants' own estimate exceeded the value shown

stance	feedback	pre_above_shown_FALSE	pre_above_shown_TRUE
Pro-choice	Better	172	28
Pro-choice	Worse	40	174
Pro-choice	Control	129	61
Pro-life	Better	143	61
Pro-life	Worse	7	198
Pro-life	Control	67	124

```
kink <- map_dfr(dvs, \(dv) imap_dfr(by_stance, \(d, s) {
  f <- lm(reformulate("feedback * moral + poly(meta_pre, 2, raw = TRUE) + ideology + above_shown", dv),
          data = d)
  tidy(f, conf.int = TRUE) |> filter(term == "above_shown") |> mutate(dv = dv, stance = s, .before = 1)
}))
show_tbl(kink, "Kink at displayed value (exploratory)")
```

Table 43: Kink at displayed value (exploratory)

dv	stance	term	estimate	std.error	statistic	p.value	conf.low	conf.high
trust	Pro-choice	above_shown	0.398	0.136	2.936	0.003	0.132	0.664
trust	Pro-life	above_shown	-0.213	0.186	-1.144	0.253	-0.578	0.152
threat	Pro-choice	above_shown	-0.309	0.147	-2.097	0.036	-0.599	-0.020
threat	Pro-life	above_shown	0.445	0.193	2.312	0.021	0.067	0.824

9. Export tables

```
out <- list(
  cell_descriptives    = cell_desc,
  manip_anova          = manip_anova,
  manip_emm            = manip_emm,
  manip_pairs          = manip_pairs,
  manip_change         = manip_change,
  manip_stance_int     = manip_int,
  prereg_anova         = prereg_anova,
  prereg_vs_control    = h_tests,
  prereg_pairs         = all_pairs,
  pt_slopes            = pt_slopes,
  pt_slope_diffs       = pt_slope_diffs,
  robustness           = robust,
  pooled_anova         = pooled_anova,
  pooled_interaction   = pooled_int,
  pooled_simple        = pooled_simple,
  kink                 = kink
)
iwalk(out, \(tbl, name) write_csv(tbl, file.path(dir_out, paste0(name, ".csv"))))
saveRDS(s4, file.path(dir_out, "s4_scored.rds"))
```

```
sessionInfo()
```

```
R version 4.6.0 (2026-04-24)
Platform: aarch64-apple-darwin23
Running under: macOS Tahoe 26.5.2

Matrix products: default
BLAS:   /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib;  LAPACK version 3.12.1

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Los_Angeles
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] knitr_1.51       broom_1.0.13     psych_2.6.5      effectsize_1.0.2
 [5] emmeans_2.0.3    car_3.1-5        carData_3.0-6    lubridate_1.9.5
 [9] forcats_1.0.1    stringr_1.6.0    dplyr_1.2.1      purrr_1.2.2
[13] readr_2.2.0      tidyr_1.3.2      tibble_3.3.1     ggplot2_4.0.3
[17] tidyverse_2.0.0  readxl_1.5.0.1

loaded via a namespace (and not attached):
 [1] generics_0.1.4     lattice_0.22-9     stringi_1.8.7      hms_1.1.4
 [5] digest_0.6.39      magrittr_2.0.5     evaluate_1.0.5     grid_4.6.0
 [9] timechange_0.4.0   estimability_2.0.0 RColorBrewer_1.1-3 mvtnorm_1.4-1
[13] fastmap_1.2.0      cellranger_1.1.0   jsonlite_2.0.0     backports_1.5.1
[17] Formula_1.2-5      scales_1.4.0       mnormt_2.1.2       abind_1.4-8
[21] cli_3.6.6          crayon_1.5.3       rlang_1.2.0        bit64_4.8.2
[25] withr_3.0.3        yaml_2.3.12        otel_0.2.0         parallel_4.6.0
[29] datawizard_1.3.1   tools_4.6.0        tzdb_0.5.0         bayestestR_0.18.1
[33] vctrs_0.7.3        R6_2.6.1           lifecycle_1.0.5    bit_4.6.0
[37] vroom_1.7.1        insight_1.5.2      pkgconfig_2.0.3    pillar_1.11.1
[41] gtable_0.3.6       glue_1.8.1         xfun_0.59          tidyselect_1.2.1
[45] parameters_0.29.2  rstudioapi_0.19.0  farver_2.1.2       nlme_3.1-169
[49] htmltools_0.5.9    labeling_0.4.3     rmarkdown_2.31     compiler_4.6.0
[53] S7_0.2.2
```